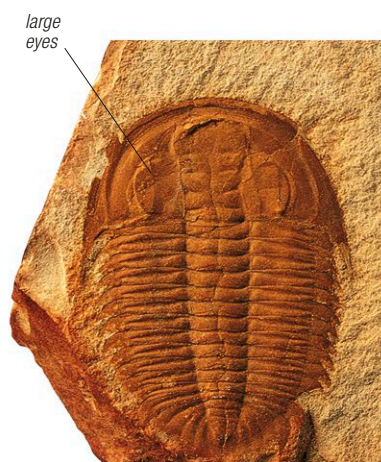


INVERTEBRATES

Invertebrates are animals that do not have a backbone. Although many of them are small—and therefore easily overlooked—they are immensely varied and widespread, accounting for about 97 percent of all known animal species. While vertebrates form a single phylum, invertebrates are an informal collection of more than 30, and the members of just one phylum—the arthropods—probably outnumber all other animals on earth. Invertebrates are found in every conceivable type of habitat, but they are most plentiful in the oceans, which is where animal life first arose.

Evolution

Exactly how multicellular invertebrate animals evolved remains uncertain, but there is little doubt that their ancestors were single-celled organisms similar to living protozoans. One theory, accepted by many biologists, is that loose colonies of these microorganisms formed more permanent partnerships

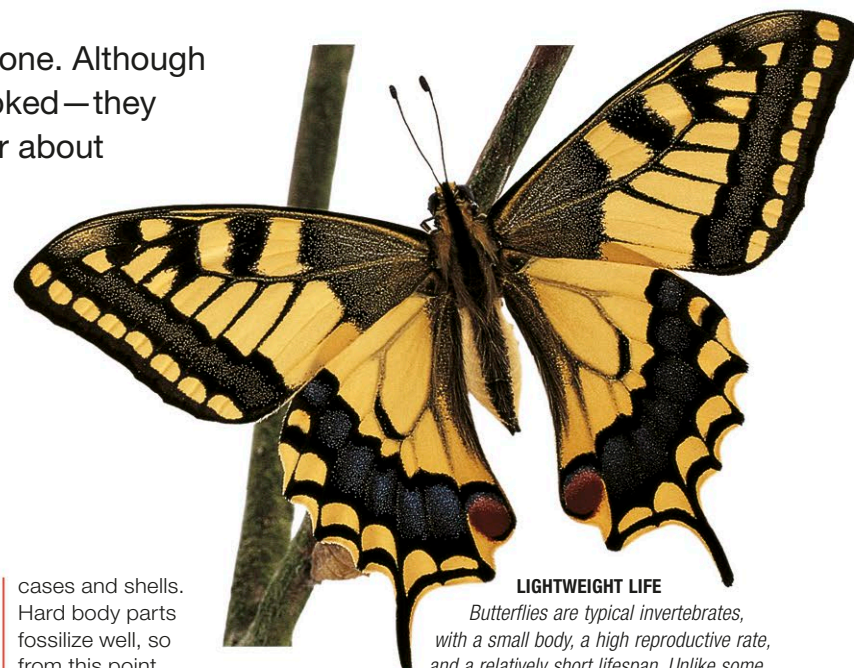


perhaps as hollow balls of cells similar to the living colonial proctotist *Volvox*. Infolding or growth of cells inside the hollow center would produce simple two-layered multi-cellular organisms—the start of invertebrate life.

The earliest signs of animal life are chemical fossils rather than the visible remains of the organisms. Carbon biomarkers specific to sponges are found in rocks dating back about 700,000 million years. The earliest fossils of invertebrates themselves come from late Precambrian times, about 600 million years ago. At the start of the Cambrian Period, about 545 million years ago, there was an acceleration of evolutionary change, known as the Cambrian Explosion. Soft-bodied animals, such as jellyfish, were joined by ones with hard body

PAST SUCCESS

Trilobites were among the most successful invertebrates that have ever lived. They existed for about 300 million years, leaving huge numbers of fossils, but died out in a mass extinction about 245 million years ago.

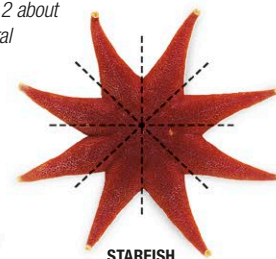


LIGHTWEIGHT LIFE

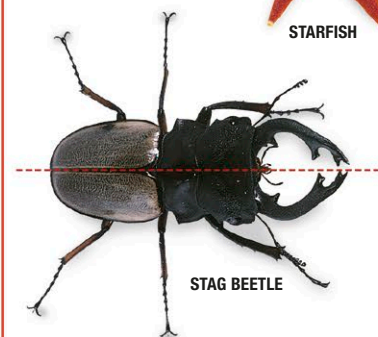
Butterflies are typical invertebrates, with a small body, a high reproductive rate, and a relatively short lifespan. Unlike some vertebrates, invertebrates are ectothermic (cold blooded), which means their activity slows down as temperatures drop.

SYMMETRY

Adult starfish appear to have radial symmetry, which means that equal body parts radiate from a central point. A stag beetle has bilateral symmetry, with a body divided in 2 about a single axis. In bilateral animals, many body parts exist in pairs.



STARFISH



STAG BEETLE

cases and shells. Hard body parts fossilize well, so from this point onward, the development of invertebrate life is much easier to trace. Although it happened long ago, the Cambrian Explosion brought into being all the invertebrate phyla that exist today. Studies of living species show that these can be divided into 2 groups: the mollusk-annelid-arthropod line and the echinoderm-chordate line. The second of these groups gave rise to animals with backbones.

Anatomy

In anatomical terms, invertebrates are united principally by things that they lack: as well as not having a backbone, they do not have a bony skeleton, and they do not have true jaws. As adults, some invertebrates look like plants, and spend their lives in one place; others are instantly recognizable as animals, because they are always on the move.

SKELETONS

Many invertebrates have a skeleton, which may be internal or external—examples of both types are found in animals that live in water and on land. Some skeletons have evolved chiefly for protection and can withstand powerful impacts, but others are more flexible and exist mainly to keep the body in shape. Skeletons are made of various materials: hard structures often consist of crystalline minerals, while the cuticle of insects is made largely of a plasticlike material called chitin. Once formed, many external skeletons cannot grow and have to be periodically shed and replaced.



SHELLS

A bivalve shell is a 2-part external skeleton. It is relatively heavy but exceptionally hard and grows in step with its owner.

SPICULE SKELETON

Most sponges have an internal skeleton of mineral slivers (spicules), set in a lattice of protein-based fibers. In this glass sponge, they are arranged with almost mathematical precision.



fluid-filled cavity

HYDROSTATIC SKELETON

The fluid-filled body cavity of an earthworm is bounded by a flexible wall. The pressurized fluid acts like the air inside a tire.



JOINTED SKELETON

Arthropods, such as this pill millipede, have an external skeleton made of hard plates. Joints between the plates make the skeleton flexible. This kind of skeleton cannot grow.